

FilaSeg: Geometry-based instance reconstruction of carbon nanotube bundles from degraded centreline masks

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[ACTION REQUIRED: add ORCIDs and confirm the official MSM affiliation wording against the current University of Twente convention]

Supervisor-review draft. This is a shortened mirror of the FilaSeg manuscript prepared for early supervisor feedback, not the submission manuscript. Several Materials-and-methods and Discussion subsections are temporarily replaced by short summaries and visible **Supervisor review note** markers; the full text lives in the canonical paper repository and will be restored after feedback.

Abstract

Converting a dense filament centreline mask into individual bundle instances is difficult: gaps break true trajectories, and crossings create several locally plausible continuations. We present FilaSeg, a deterministic geometry-based method that takes a binary carbon-nanotube (CNT) bundle-axis mask and separates it into individual bundle instances through staged, topology-checked geometric reconnection; a final stage then uses the paired scanning electron microscopy (SEM) image to estimate each instance’s apparent width and render it. Methods are compared on a common, method-independent partition of the input skeleton into fragments, using pairwise grouping F_1 on equivalent per-instance centreline representations, evaluated before width rendering. After the evaluation configuration was frozen, FilaSeg and a minimum-turn skeleton tracer were evaluated on 125 independently seeded procedural scenes. FilaSeg achieved mean common-fragment $F_1=0.734$ (density-stratified 95% bootstrap CI [0.722, 0.745]), with a paired FilaSeg-minus-minimum-turn difference of 0.071 (95% CI [0.059, 0.082]). This advantage was not uniform: it held at four of the five tested densities, and its interval crossed zero at the fifth. Scoring the rendered Stage-4 output instead of the Stage-3 centreline gave a lower score, showing that grouping accuracy and rendering accuracy must be assessed separately. A single manually annotated SEM crop was used during development and is not a held-out test. Taken together, this predefined synthetic evaluation supports a conditional advantage over the fixed minimum-turn comparator for the tested generator; it does not establish general superiority over filament-tracing methods, or transfer to other SEM instruments, materials, or upstream segmenters.

Supervisor review note: A broader upstream pipeline also exists around the method described in this manuscript: real SEM images and procedurally generated masks → CycleGAN image translation → U-Net semantic segmentation → binary CNT bundle-axis mask → FilaSeg instance extraction → SEM-assisted Stage-4 width estimation, overlap handling, and rendering. This manuscript currently treats the binary axis mask as the formal input and evaluates only the instance-extraction stages. Open questions for supervisor feedback:

- Should the complete upstream CycleGAN/U-Net pipeline be integrated into this paper?
- Should it instead remain a separate paper or companion workflow?
- Should the current manuscript continue to treat the binary axis mask as the formal input?

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- How much of Stage 4 should be presented as part of instance extraction versus a separate SEM-assisted measurement/postprocessing stage?

Keywords: carbon nanotubes; scanning electron microscopy; instance segmentation; curvilinear structure extraction; Hough transform; filament tracing; image-based microstructural characterization.

1 Introduction

The macroscopic properties of carbon nanotube (CNT) films, yarns and composites are governed by the geometry of the bundle network rather than by the properties of individual nanotubes alone: the length distribution of the bundles, their width, their local orientation and alignment, their curvature, and the density and topology of the crossings between them all enter the mechanical and electrical response [1, 2]. Mesoscopic and network models of CNT assemblies take exactly this description, slender mutually contacting elements with measurable geometry, as their input [3–5]. Scanning electron microscopy (SEM) is the routine imaging modality for such materials, yet the quantitative analysis that follows an acquisition usually stops at bulk descriptors: an areal coverage fraction, a mean width, or a global orientation-order parameter. These summarize the network but discard the object that quantitative characterization actually needs, the individual bundle as an addressable instance with its own centreline, width, endpoints and crossings. Image analysis of CNT micrographs has followed the same emphasis, concentrating on alignment and orientation, quantifying nematic order or the orientation distribution of CNT forests and films [6, 7]. Recent work has also applied neural networks to identify CNT foreground in microscopy images [8, 9], while broader reviews summarize deep-learning segmentation across nanomaterial imaging [10].

The obstacle is not semantic segmentation. Convolutional segmentation networks such as U-Net [11] and its self-configuring successors [12] produce good foreground masks of CNT structures. Instance-level extraction, partitioning that foreground into the correct set of individual bundles, remains challenging in dense, overlapping filament networks, for four coupled reasons. First, bundles are effectively one-dimensional objects embedded in a two-dimensional image, so a single faint patch a few pixels long splits one bundle into two mask fragments. Second, bundles cross and touch, so foreground connectivity does not correspond to bundle identity: two crossing bundles form one connected component that must be split, while one faded bundle forms two components that must be joined. Third, the bundles are densely packed, so at a crossing a locally correct continuation and a locally plausible wrong continuation are often geometrically indistinguishable. Fourth, the objects are only a few pixels wide, so pixel-overlap instance metrics and one-pixel skeleton artefacts dominate naive approaches. Standard instance-segmentation priors assume a compact object with a clear interior and boundary: a bounded blob, a flow field pointing toward a centre, or a star-convex shape [13–15]. CNT bundles do not fit that assumption. They are long, thin, crossing curves with no meaningful centre, and they may overlap.

Four classes of existing method address parts of this problem. *Ridge and tubular-structure detectors* [16–18] recover sub-pixel line positions and enhance line-like structure from grayscale intensity, but they do not assign pixels to individual objects across a junction. *Fibre-network tracing tools* developed for collagen and biopolymer networks, including FIRE and CT-FIRE [19, 20], SOAX [21] and the FiNTA tracing algorithm [22], return per-fibre length, width and orientation, but they target sparser, often micron-scale fibres and yield a fibre graph rather than a set of separately rendered, individually addressable instances. *Aggregate measurement tools* such as DiameterJ [23] deliver well-validated width and pore-size distributions over a whole micrograph, but as image-level distributions rather than per-instance geometry, and they do not resolve crossing ownership; the individual-fibre separation approach of Depriester et al.

[24] addresses that separation step directly for fibrous materials. *Learned instance methods* for filaments are emerging: FibeR-CNN [25] extends Mask R-CNN to image-based fibre analysis, including CNT-like fibres, as an independent raw-image detector rather than a centreline-mask grouping method, and Pick and Trace [26] segments filamentous instances with a recurrent network. Methodologically closest to the present reconnection stage is the terminus-pairing approach of Liu et al. [27], which separates a raw microscopy image into orientation-specific branches with a learned network and then pairs terminus points across branches to regroup fragments into individual filaments; it was validated on microtubule micrographs rather than on a precomputed centreline mask.

Supervisor review note: Liu et al. [27] is included here as the closest published methodological analog to FilaSeg’s own orientation-binned vectorization and tip-pairing reconnection (orientation separation, then pairing of fragment endpoints). This repository has no internal record of FilaSeg’s design being directly derived from that work; please confirm before this is described as a direct methodological predecessor rather than an independently convergent design.

Instance tracing of crossing curvilinear objects is also mature in other domains, in neuron reconstruction [28–30], plant-root architecture [31] and crack segmentation [32], but each embeds a domain-specific prior that dense CNT bundle films or networks (referred to here as *CNT mats*) do not satisfy: bundles are open curves that cross freely and carry neither a root nor a hierarchy.

The closest published work is that of Peters et al. [33], who combine U-Net-like semantic segmentation with algorithmic fibre tracing, crossing separation and width refinement from transverse grayscale profiles, and validate on SEM images of CNT materials for aerosol characterization and exposure assessment. Two-stage semantic-segmentation-plus-tracing designs therefore already exist for CNT SEM, and the contribution claimed here is not the two-stage idea itself. What distinguishes the present work is the operating regime and the representation. Peters et al. [33] target low-occupancy aerosol-filter images, in which fibres are comparatively isolated; the target here is a dense CNT mat in which the crossing density, not the detectability of a single fibre, sets the difficulty. The input representation differs accordingly: FilaSeg consumes a thin-axis centreline mask rather than a full bundle silhouette, and emits a multi-label representation in which overlapping instances retain their shared pixels at a crossing instead of one instance claiming them. The method differs in using tiled Hough-based residual vectorization followed by staged, topology-checked reconnection, and the evaluation differs in reporting a pair-weighted grouping diagnostic that separates over-merging from under-merging, in stress-testing across a controlled range of crossing density, and in adapting every pixel-scale parameter to the physical pixel size. These are differences in design, not a head-to-head performance comparison with Peters et al. [33] or the other related methods above; the present evidence does not establish quantitative superiority over them (Section 4.3).

This paper presents FilaSeg, a deterministic, geometry-based postprocessing method that converts a binary filament-centreline mask and its paired SEM image into separated filament instances. The input is a binary centreline mask together with the grayscale SEM image from which it was derived; the output is a set of possibly overlapping instances, each carrying a pixel support, an ordered centreline, a width, a length and endpoints, stored as a multi-label stack so that crossings are preserved. These geometric fields are exported for downstream use but are not claimed as validated measurements in the present study. The method proceeds in four stages: tiled Hough-based residual vectorization, orientation-aware fragment cleaning, geometry-based instance reconnection, and SEM-gradient-based width estimation with final rendering. It is evaluated on one manually annotated real SEM crop and on a procedural synthetic benchmark that renders thick bundles for the SEM channel while feeding the method the thin axis that an upstream segmenter actually reconstructs. Three questions are addressed: how grouping accuracy depends on the generator density; whether the conclusion changes when methods are scored on equivalent centreline representations; and which scenes favor FilaSeg or a minimum-turn comparator. The semantic segmentation that produces the input mask is treated

throughout as an external upstream process and is not evaluated here. The real-data evidence rests on a single annotated development crop, so it is presented qualitatively rather than as a population estimate.

2 Materials and methods

2.1 CNT material and SEM imaging

The real micrographs analysed here are grayscale secondary-electron SEM images of carbon nanotube bundle mats. The reference acquisition has a horizontal field width of $1.66\text{ }\mu\text{m}$; the instrument metadata reports this field width over 1536 px, giving a physical pixel size of $u_{\text{ref}} = 1.66/1536 \approx 1.081 \times 10^{-3}\text{ }\mu\text{m px}^{-1}$ ($\approx 1.08\text{ nm/px}$), and the stored micrographs measure $1534 \times 1026\text{ px}$ after removal of the instrument data bar. Images are 8-bit grayscale and are used without intensity preprocessing: all intensity-dependent quantities in Section 2.7 are evaluated on the raw 8-bit scale $S \in [0, 255]$. Instance analysis is performed on 512×512 crops taken on the 128-px tile grid used by Section 2.4, so that tile boundaries are identical between a crop and its parent image.

[MISSING INFORMATION: SEM instrument, detector, accelerating voltage, working distance, specimen preparation, CNT material identity, and acquisition settings are currently unavailable and must be added before submission.]

The primary quantitative evaluation in this paper is synthetic (Section 2.8.2) and does not depend on these acquisition parameters. They are nonetheless required before submission because the manuscript shows a real SEM image, reports a numerical result on a manually annotated real crop, uses SEM intensity for apparent projected-width estimation, and is positioned as applicable to SEM-derived centreline masks. Until they are supplied, the real-data result is a limited development case study and not a general estimate of performance across specimens, microscopes, imaging conditions or CNT materials (Section 4.3).

2.2 Input masks, problem definition and mask provenance

Problem definition. The input to FilaSeg is a pair: a binary centreline mask $M \in \{0, 1\}^{H \times W}$, in which the foreground marks the *axis* of each filament rather than its full silhouette, together with the paired grayscale SEM image $S \in [0, 255]^{H \times W}$ from which M was derived. The mask is a thin ribbon, nominally one to three pixels wide, and crossings are represented as shared foreground pixels: where two filaments cross, the mask carries a single undifferentiated foreground region and gives no indication of which pixels belong to which filament. The required output is an instance-labelled mask: a set of filaments $\{B_1, \dots, B_K\}$, each with a pixel support Ω_k , an ordered centreline $\mathcal{C}_k$, a scalar apparent width w_k , a length ℓ_k and endpoints, such that each B_k corresponds to one physical bundle even where bundles cross or where the mask is locally broken. Because two crossing bundles genuinely share pixels, output instances are permitted to overlap: the labelling is stored as a multi-label stack rather than a single-valued label image, so that a disputed crossing pixel can be claimed by both instances instead of being awarded arbitrarily to one.

What is and is not evaluated. FilaSeg is evaluated primarily as a centreline-mask-to-instance method, using synthetic masks with exact instance ground truth (Section 2.8.2). One real SEM crop is included as a manually annotated development case study (Section 2.8.1). The upstream method used to produce that crop’s centreline mask is not evaluated in this study. Throughout, *exact ground truth* refers only to synthetic data, where the generator defines each instance by construction, and *manual reference annotation* refers to the real crop, where the reference is one human labelling of an ambiguous image rather than an unquestionable truth.

Provenance of the real masks. The centreline masks used in the real case study were generated by a separate synthetic-data-driven CycleGAN/U-Net workflow [34], itself adapted from the open particle-segmentation workflow of Rühle et al. [35]. That upstream workflow is not evaluated in this study and is treated only as one possible source of FilaSeg input; any segmenter producing a thin-axis binary mask can be substituted. Its one property that matters here is that it reconstructs the bundle *axis* rather than the full bundle silhouette, which is why the input mask is thin and why the physical width must be recovered separately from the SEM image (Section 2.7).

The training-set relationship between the upstream semantic segmenter and the real reference crop could not be reconstructed from the available records: the segmenter was developed in a separate repository, and no training-image list, data-split record or training log for it is present in the FilaSeg repository. The real crop is therefore treated only as a development case study, and no independent end-to-end SEM generalization claim is made. Because the primary evaluation of this paper is synthetic, and its input masks are generated procedurally rather than by any trained model (Section 2.8.2), this gap does not affect the synthetic instance-grouping experiments; it bounds only the interpretation of the single real crop.

2.3 FilaSeg overview and physical-scale adaptation

FilaSeg factorizes the mask-to-instances problem into four ordered stages (Figure 1):

1. **Hough-based vectorization** (Section 2.4): decompose M into short, oriented line primitives, bridging small raster gaps in the process.
2. **Orientation-aware fragment cleaning** (Section 2.5): remove specks, close along-filament gaps with orientation-matched morphology, and reduce each primitive to a single smooth two-tip curve.
3. **Geometry-based instance reconnection** (Section 2.6): merge fragments into instances with a staged, topology-checked greedy graph merge governed by geometric gates.
4. **SEM-gradient-based width estimation and rendering** (Section 2.7): measure one apparent width per instance from the SEM intensity profile across the centreline, render each instance at that width, and resolve residual overlaps.

The corresponding implementation modules are named `stringart`, `preprocess`, `reconnect` and `postprocess` in the released code; the scientific names above are used throughout the manuscript.

Physical-scale adaptation. All pixel-valued defaults are tuned at the reference pixel size u_{ref} of Section 2.1 (the physical size of one pixel in the reference acquisition). Two mechanisms let one configuration transfer to other magnifications. First, for an image whose physical pixel size is u (the physical size of one pixel in that image), define the dimensionless pixel-scaling factor

$$s_{\text{px}} = \frac{u_{\text{ref}}}{u}, \quad (1)$$

so that $s_{\text{px}} < 1$ at a coarser pixel size. Every configured parameter is then rescaled according to its physical dimension,

$$d_{\text{px}}(u) = s_{\text{px}} d_{\text{px,ref}}, \quad A_{\text{px}}(u) = s_{\text{px}}^2 A_{\text{px,ref}}, \quad \kappa_{\text{px}}(u) = \kappa_{\text{px,ref}} / s_{\text{px}}, \quad (2)$$

where $d_{\text{px,ref}}$ is any length-valued reference parameter in pixels, $A_{\text{px,ref}}$ any area-valued reference parameter in pixels squared, and $\kappa_{\text{px,ref}}$ the reference curvature threshold in inverse pixels; $d_{\text{px}}(u)$, $A_{\text{px}}(u)$ and $\kappa_{\text{px}}(u)$ are their scaled counterparts at pixel size u . Dimensionless parameters

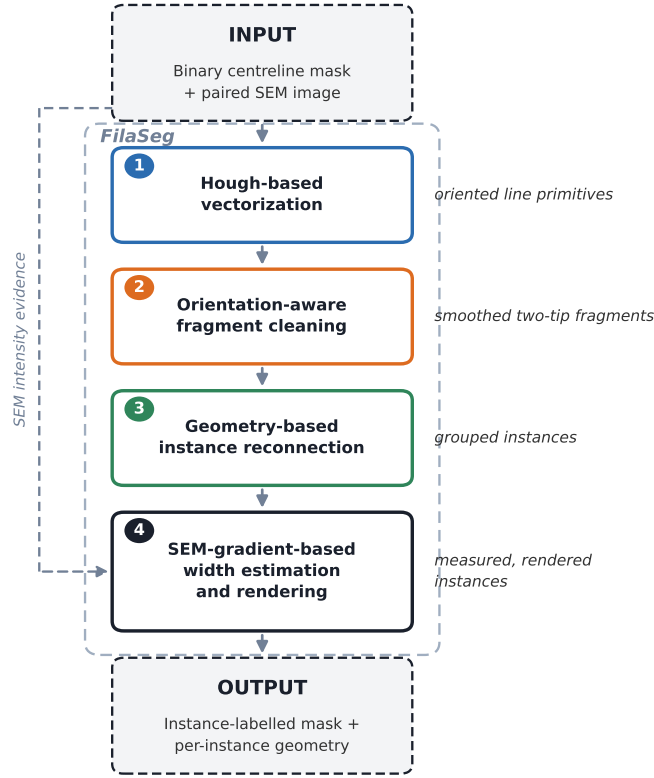


Figure 1. Scope of FilaSeg. The method consumes a binary centreline mask and its paired SEM image and returns an instance-labelled mask with per-instance geometry. The four stages operate on geometry alone except for the final width estimation, which reads the SEM intensity across each centreline. The semantic segmentation that produces the input mask is external to the method and is not evaluated in this study.

(intensity thresholds, cosines, fractions, iteration counts) are left unchanged. The physical pixel size is supplied explicitly or obtained from available SEM acquisition metadata; the reference scale is used only as a documented fallback. A scaled copy of the configuration is written per run so that the source configuration is never mutated. Second, within the vectorization stage, four parameters adapt to the image *content* rather than to its nominal scale, as described in Section 2.4.

Two conventions are fixed here and used throughout. Filament axes are unoriented, so the mismatch between two axis angles θ_1, θ_2 is the circular difference

$$\Delta\theta = \min(|\theta_1 - \theta_2|, 180^\circ - |\theta_1 - \theta_2|) \in [0^\circ, 90^\circ]. \quad (3)$$

Second, the transverse size of a bundle is reported as an *apparent projected width*: the SEM records a projected silhouette, and no assumption is made that the bundle cross-section is circular, so the term *diameter* is avoided.

2.4 Stage 1: tiled Hough-based residual vectorization

Stage 1 (implementation module `stringart`) converts the input mask into a set of oriented line primitives. It applies a tiled, orientation-binned residual decomposition built on the progressive probabilistic Hough transform [36]: within each image tile and orientation bin, it repeatedly accepts the line segment that explains the most currently unexplained mask pixels, which also bridges small raster gaps along the way. Four of its parameters adapt automatically to the

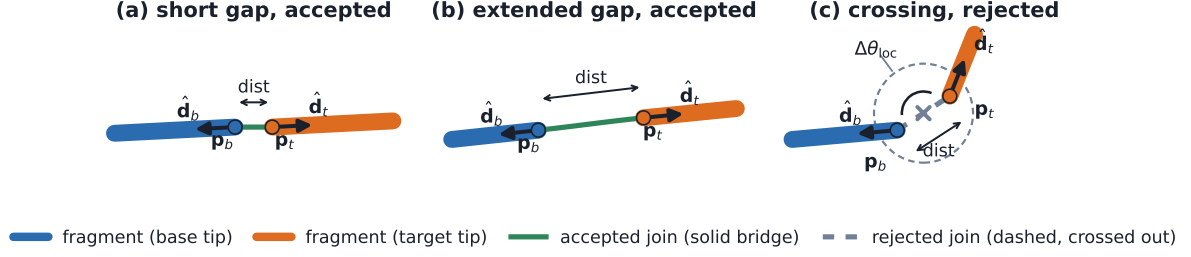


Figure 2. Geometry of a candidate reconnection and three representative outcomes: (a) a short, near-collinear gap accepted by the short-gap pass; (b) a longer gap accepted because the bridged trace remains smooth; (c) two fragments meeting at a steep angle rejected because their local principal orientations disagree, even though the tips point at each other. Threshold values are omitted from this simplified draft.

observed filament width rather than to a fixed pixel count, so the same configuration transfers across the physical scales of Section 2.3.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

2.5 Stage 2: orientation-aware fragment cleaning

Stage 2 (**preprocess**) cleans the per-orientation primitives produced by Stage 1: it discards specks below a minimum size, closes short along-filament gaps with orientation-matched morphology, and reduces each fragment to a single smooth two-tip curve. This hands Stage 3 physically plausible curves with well-defined tip geometry, rather than fragmented masks whose tip tangents are dominated by raster noise.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

2.6 Stage 3: geometry-based instance reconnection

Stage 3 (**reconnect**) merges fragments into filament instances. It represents the current segmentation as a set of components with skeleton tips, and greedily joins compatible tip pairs whose local geometry, distance, collinearity, curvature and smoothness all pass a battery of gates, including a local-orientation crossing-prevention test that targets the characteristic failure mode of fusing two filaments that merely cross (Figure 2). Thresholds are applied in three ordered passes, from safest to broadest, and a topology-checked repair step recovers configurations that simple endpoint-to-endpoint projection misses.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

2.7 Stage 4: SEM-gradient-based width estimation and rendering

Stage 4 (**postprocess**) converts reconnected labels into measured, rendered instances. For each surviving instance it measures one apparent projected width from the paired SEM intensity profile sampled across the centreline, using the fact that a bundle appears as a band bounded by two opposite-polarity intensity edges; it then renders the instance at that width and resolves residual overlaps between instances.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

Thick-bundle / thin-axis synthetic scenes with exact ground truth

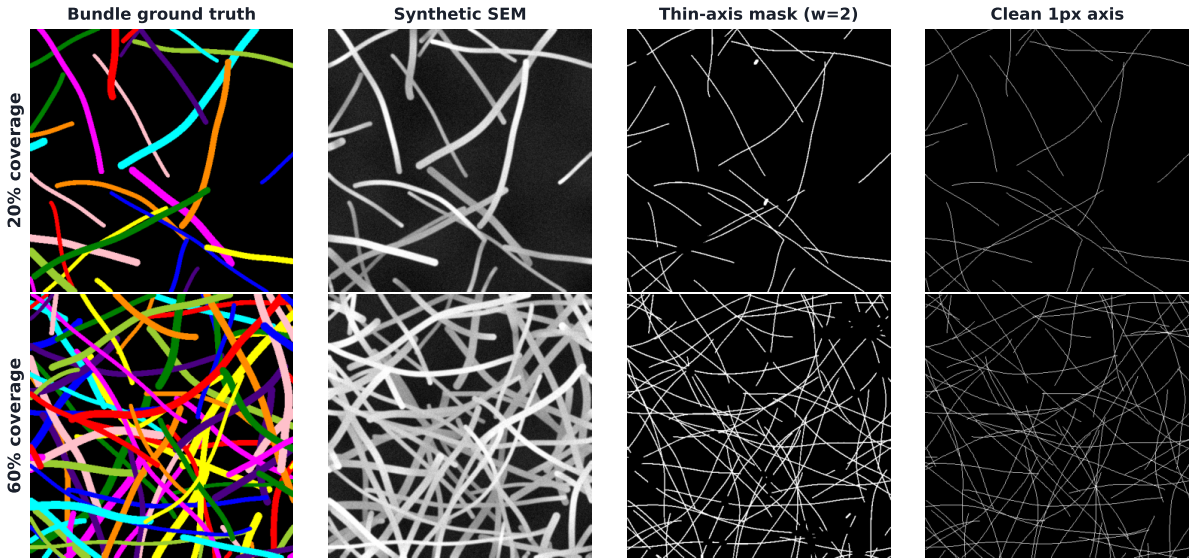


Figure 3. Procedural synthetic benchmark scenes at two areal densities. Each row shows the thick-bundle instance ground truth, the synthetic SEM image, the degraded thin-axis mask fed to FilaSeg, and the zero-degradation one-pixel clean axis used as a topology-only reference.

2.8 Data and reference annotations

2.8.1 Real test crop

One manually annotated instance ground truth is available: a 512×512 crop of a parent micrograph, containing 58 labelled filament instances. It informed method development and is used only as a qualitative case study, not a held-out population estimate (Section 4.3).

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

2.8.2 Procedural synthetic benchmark

Manual instance annotation of a dense CNT mat is expensive, so exact ground truth at scale comes from a procedural synthetic generator. Each filament is rendered as a thick band with an overlap-aware instance ground truth, and the mask FilaSeg actually consumes is a degraded thin-axis ridge of configurable width, matching the axis reconstruction an upstream segmenter would produce. The benchmark spans five prescribed areal coverages from 20% to 60%.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

2.9 Evaluation metrics and statistical analysis

Methods are compared on a common, method-independent partition of the input skeleton into atomic fragments, so that FilaSeg and every baseline are scored on one fixed partitioning problem regardless of instance numbering.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

2.9.1 Primary metric: common-fragment pairwise F_1

The primary endpoint is pairwise grouping F_1 over pairs of common fragments: precision penalizes fragment pairs from different filaments that were merged (over-merging), and recall penalizes fragment pairs from the same filament that were left separate (under-merging). The locked primary comparison reduces every method’s output to an equivalent one-pixel centreline before scoring, so that grouping accuracy is assessed separately from Stage-4 width rendering.

2.9.2 Object-level instance recovery

A count-based companion to the pairwise F_1 classifies each ground-truth filament as well recovered, deleted (absorbed into another instance), or fragmented (split across several predicted instances), and each predicted instance with no dominant ground-truth match as false.

2.10 Configuration locking, baselines and ablation protocol

All synthetic and real development data were used to tune the method and select baseline operating points before the FilaSeg configuration and baseline parameters were frozen; a disjoint set of previously unevaluated geometries was then scored once, with no further tuning. The connected-component sanity check labels every connected component of the mask with no gap reconnection or junction resolution; the main comparator skeletonizes the mask and greedily continues branches through junctions or gaps by minimum turning angle, with only its maximum gap and maximum turn selected from development data.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

2.11 Implementation

FilaSeg is implemented in Python on the standard scientific stack (NumPy, SciPy, scikit-image, OpenCV, Matplotlib, tifffile, PyYAML), as four separate stage modules orchestrated by a single end-to-end runner. The commands that reproduce every number in this paper are listed in Appendix C.

3 Results

The primary predefined comparison uses the method-independent common fragments and equivalent one-pixel instance centrelines defined in Section 2.9.1. Overall intervals use density-stratified scene resampling, and method differences are paired by scene. Stage-4 rendering and the connected-component sanity check are secondary.

Supervisor review note: The subsection numbering and figure/table assignments below follow the content of the current restructured (Pipeline-2B-only) manuscript rather than the section-by-section list in the mirror task brief: two items in that brief (a curvature/ $q_{\max}$ development study, and a component-ablation table) describe material that currently lives in the Appendix, not in this chapter, and one item (a figure with x positions at 1, 2 and 3 pixels) matches the mask-width sensitivity figure in Section 3.4 rather than the section the brief associated it with. Please confirm this mapping is what was intended.

3.1 Locked synthetic grouping comparison

FilaSeg Stage 3 achieved mean common-fragment $F_1=0.734$ (95% CI [0.722, 0.745]) on 125 independently seeded scenes, against 0.749 mean precision and 0.724 mean recall. Table 1 and Figure 4 summarize the comparison against the fixed minimum-turn tracer.

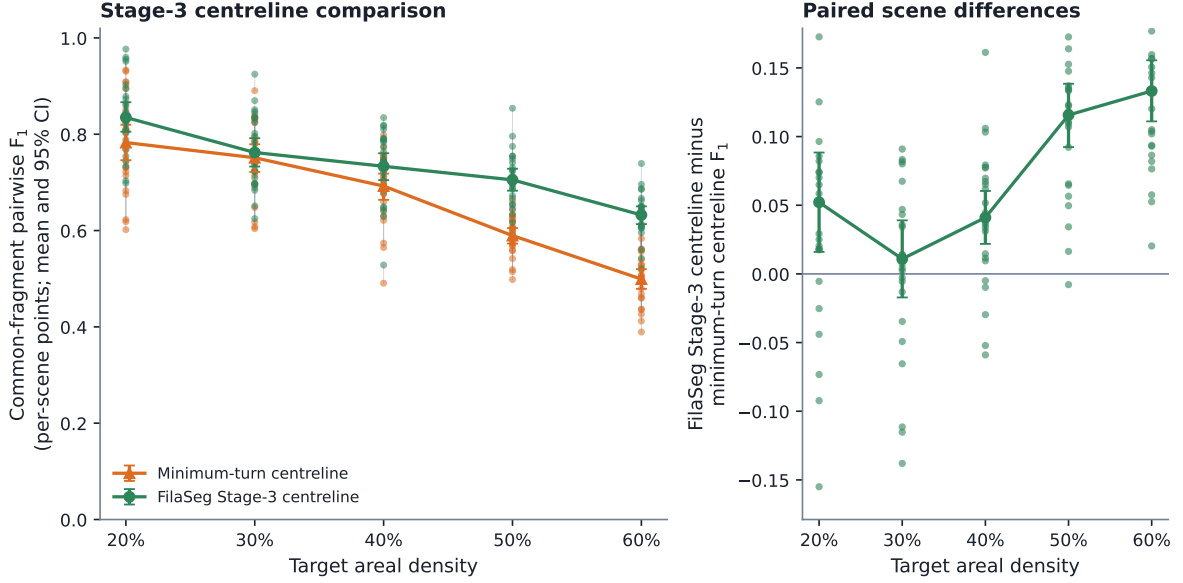


Figure 4. Common-fragment F_1 on equivalent one-pixel instance centrelines, FilaSeg Stage-3 versus the fixed minimum-turn tracer only (the connected-component sanity check is reported separately in the Supplementary Information). The left panel shows every scene plotted at its exact prescribed density, with density means and 95% bootstrap intervals and no horizontal displacement. The right panel shows the paired FilaSeg-minus-minimum-turn difference; intervals are paired within scene.

Supervisor review note: Detailed methods text intentionally omitted from this simplified review draft.

Table 1. Locked grouping comparison on equivalent per-instance centrelines representations. Values are scene means, with density-stratified 95% bootstrap intervals for F_1 and the paired FilaSeg-minus-minimum-turn contrast. “Frag. recovery” is the object-level companion defined in Section 2.9.2.

Method	P	R	F_1	Frag. recovery
Minimum-turn skeleton	0.725 [0.712, 0.738]	0.616 [0.602, 0.631]	0.663 [0.651, 0.675]	0.513 [0.497, 0.528]
FilaSeg (Stage 3 centrelines)	0.749 [0.735, 0.763]	0.724 [0.710, 0.737]	0.734 [0.722, 0.745]	0.591 [0.575, 0.607]
Paired difference (FilaSeg – minimum-turn)	0.024 [0.009, 0.038]	0.107 [0.094, 0.120]	0.071 [0.059, 0.082]	0.079 [0.062, 0.096]

The paired FilaSeg-minus-minimum-turn difference, 0.071 (95% CI [0.059, 0.082]), excludes zero in aggregate but crossed zero at the 30% density level, so this predefined evaluation supports a conditional advantage over this comparator, not a general superiority claim.

3.2 Representation audit

The post hoc evaluator audit changed neither method outputs nor the locked configuration. It reproduced the stored legacy Stage-4 F_1 values exactly, then rescored equivalent representations with ground-truth-unassigned common fragments retained as singletons. Reducing the Stage-4 output to centrelines gave mean $F_1=0.725$, whereas the rendered Stage-4 masks gave 0.681. Relative to the Stage-3 centrelines endpoint, the paired changes were -0.009 and -0.053, respectively. These are two negative representation diagnostics, not an exact mechanistic decomposition: skeletonizing rendered instance layers can shift their centrelines. The rendered output is scientifically relevant to downstream use, but it is not a representation-neutral grouping endpoint.

3.3 Locked qualitative examples

Examples were selected automatically from the corrected per-scene paired differences: the three largest FilaSeg wins, the three largest minimum-turn wins, and the two closest ties. No image was selected by visual inspection. Within each method panel, maximum-total-IoU one-to-one assignment gives a matched prediction the color of its ground-truth instance. Unmatched predictions are magenta, and missed ground-truth instances have dashed yellow outlines. The outcome panel scores the fixed common fragments and therefore shows which method grouped each item correctly.

Largest FilaSeg wins

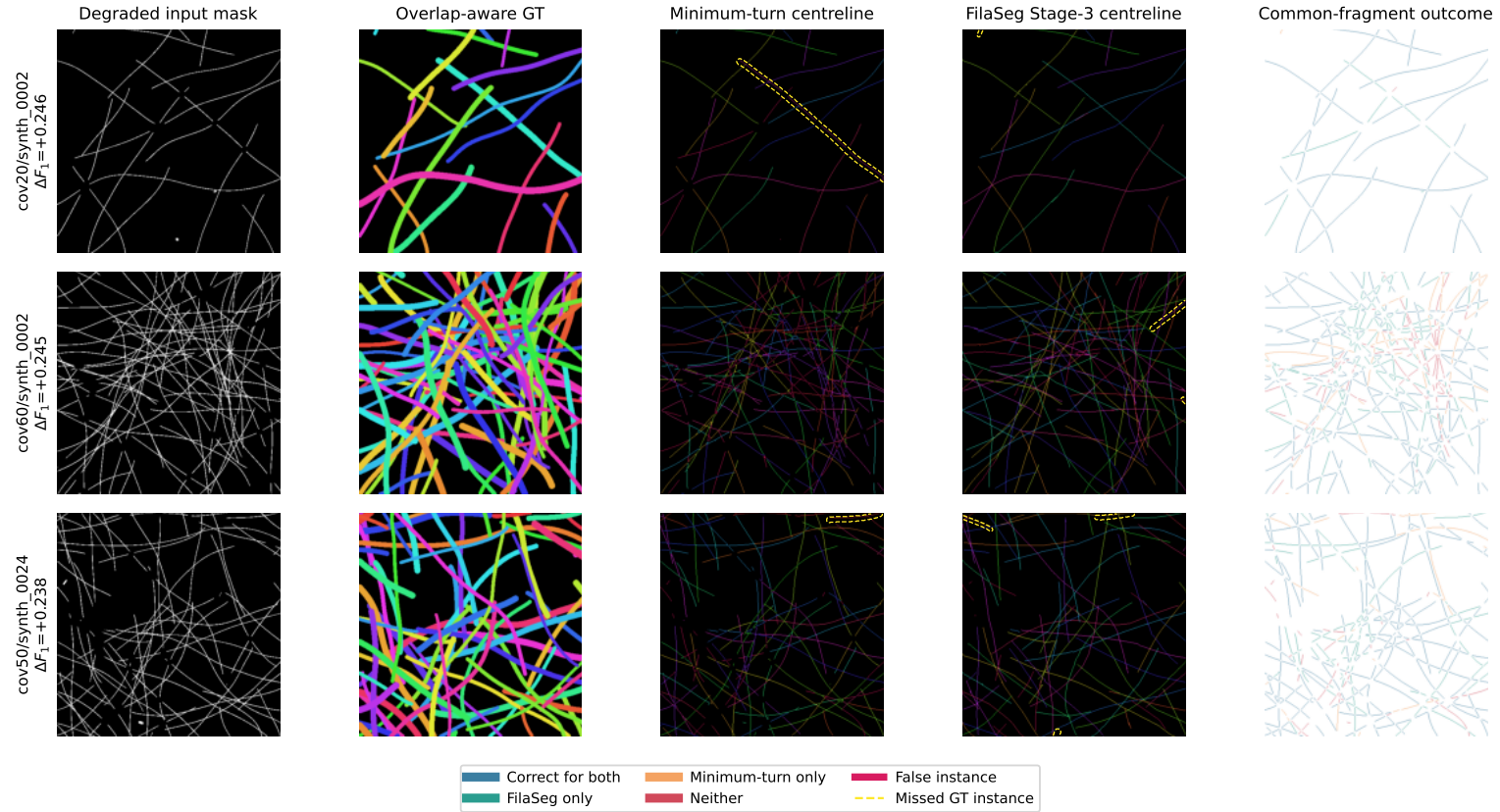


Figure 5. The three largest locked FilaSeg wins under the corrected primary endpoint. Rows and paired differences were selected by the stored evaluator rule, not by visual preference.

Largest minimum-turn wins

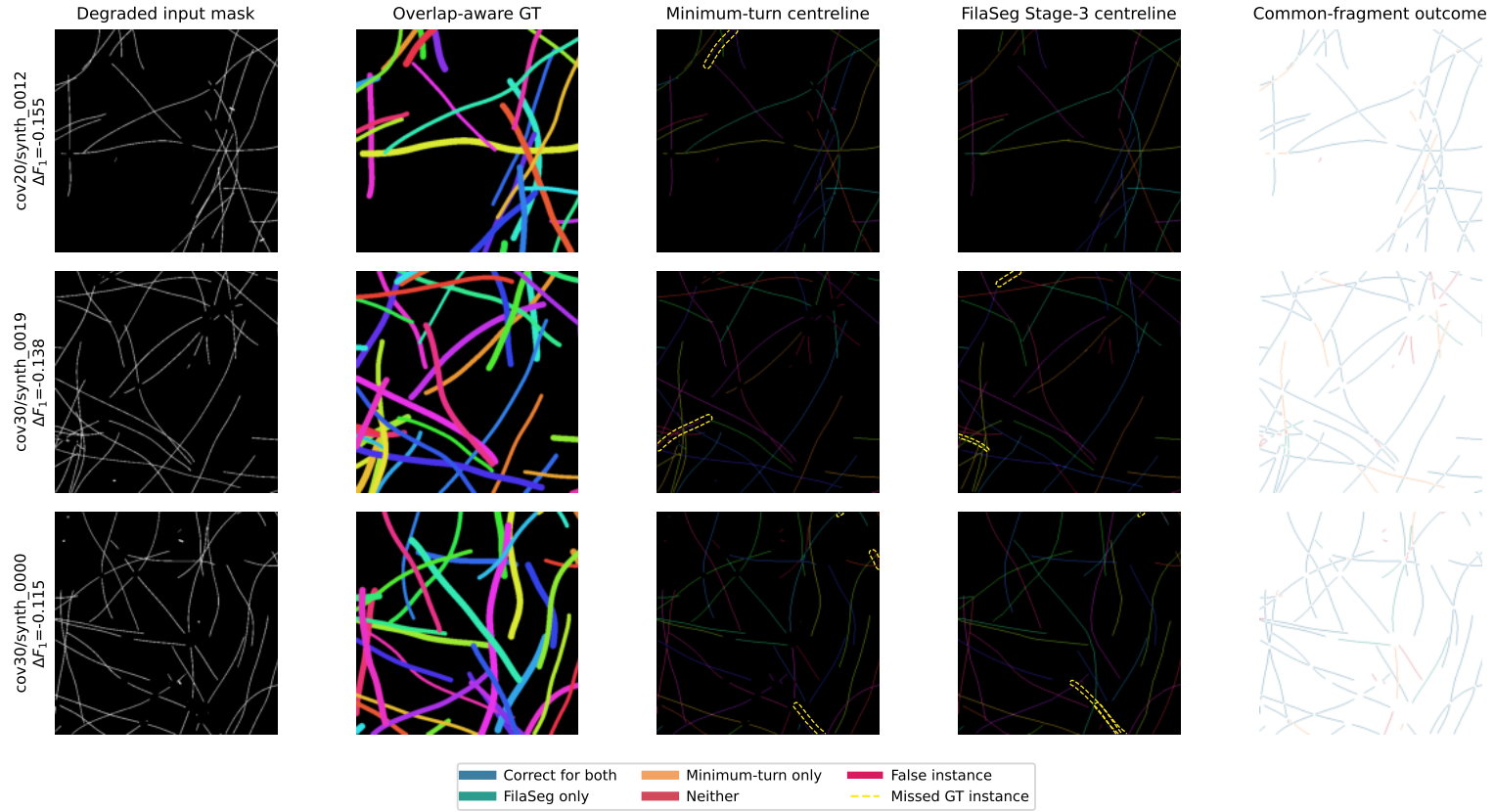


Figure 6. The three largest locked minimum-turn wins. These counterexamples remain in the main paper to show where local minimum-turn continuation was more accurate than the FilaSeg gate battery.

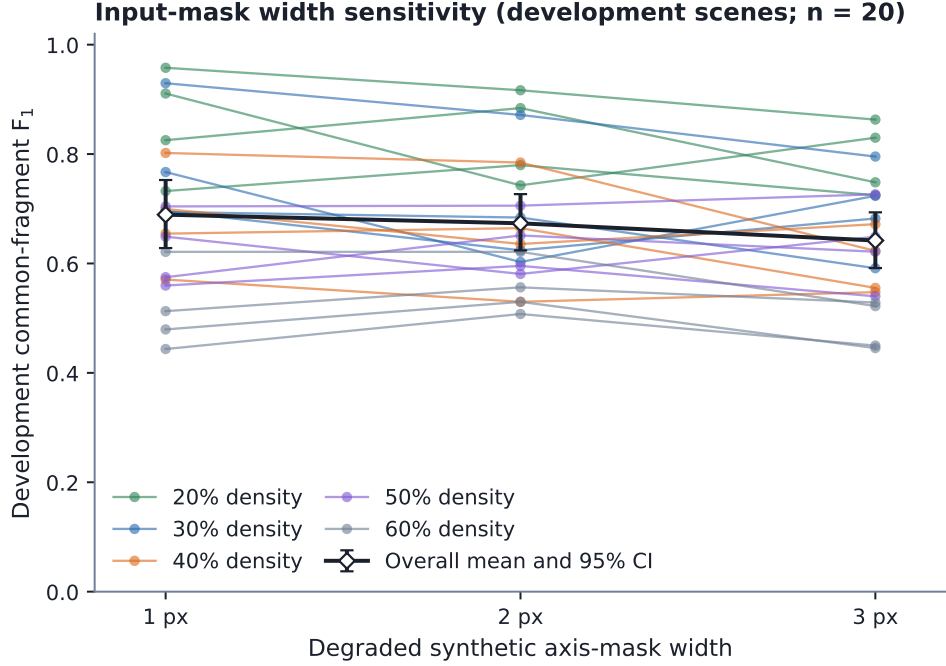


Figure 7. Development-only sensitivity to axis-mask width. Thin lines connect each scene across the three mask widths. Larger symbols show density-stratified means and 95% bootstrap intervals.

3.4 Development sensitivity studies

Input-mask width was assessed on the 20-scene development set, not on the locked benchmark. The one-pixel mask had the highest mean common F_1 . The paired interval for the two-pixel mask included zero, whereas the three-pixel mask showed a resolved decrease. Thus the tested thin representation was preferable, but the two-pixel comparison did not resolve a difference at the 95% level.

Replacing the degraded one-pixel masks by their clean procedural axes produced a small negative mean change whose paired interval included zero. At 30% coverage, all four available development scenes decreased, but the sample is too small for a density-specific inference. Under this generator and configuration, removing the modeled degradation did not produce a resolved aggregate improvement. This is a conditional diagnostic, not evidence that real segmentation errors are unimportant.

3.5 Real SEM development case study

The single 512×512 real crop contains 58 manually annotated filament instances. It informed development and therefore provides neither a held-out estimate nor a population confidence interval; every real-data result in this paper comes from this one development crop (Section 4.3). Figure 9 is descriptive: predicted instances inherit the color of their one-to-one matched manual instance, false predictions are magenta, and the one unmatched manual instance is outlined in yellow. The final rendered instances are shown because this panel illustrates the delivered output, not because the thick representation is the primary grouping endpoint.

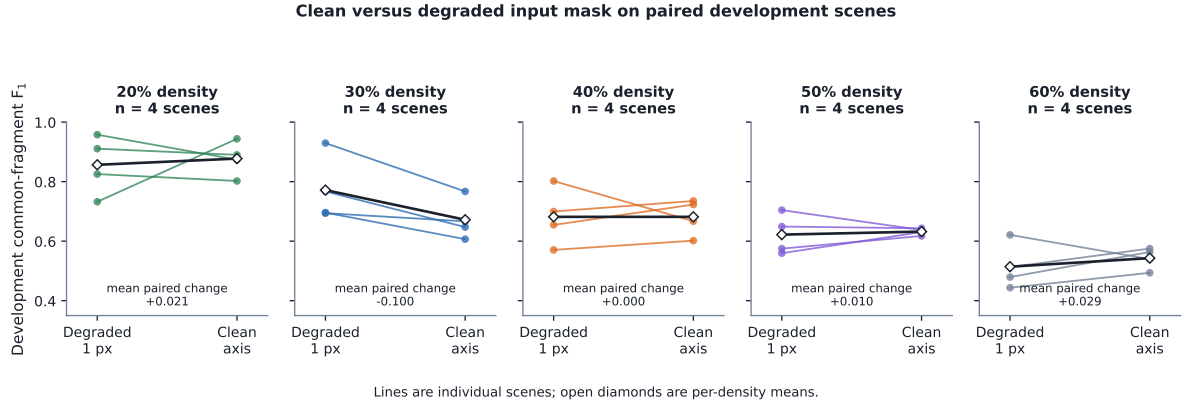


Figure 8. Development-only clean-versus-degraded comparison. Each density panel contains the four paired scenes available at that level; the small per-density sample is shown explicitly rather than summarized by an unpaired bar.

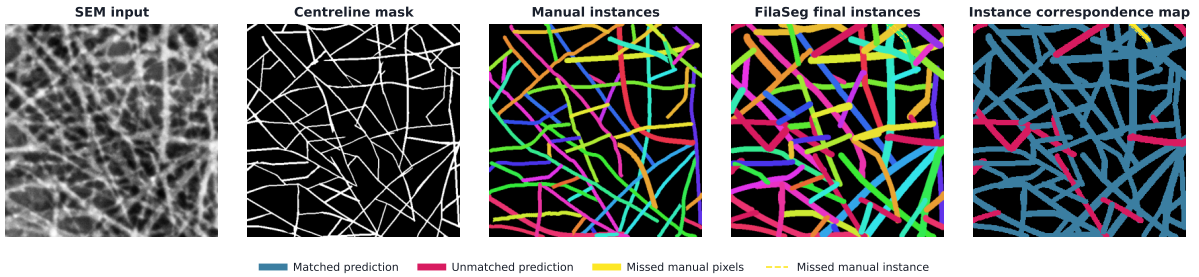


Figure 9. Real development crop. Corresponding manual and predicted instances use the same colors after maximum-total-IoU one-to-one assignment. Magenta marks unmatched predictions and dashed yellow marks the missed manual instance. The final panel summarizes instance correspondence without treating width disagreement as a grouping error.

4 Discussion

4.1 What the locked comparison establishes

The predefined benchmark isolates one question: given the same degraded one-pixel centreline mask, how well do FilaSeg Stage 3 and a fixed minimum-turn tracer group a method-independent set of fragments? FilaSeg’s paired advantage over the tracer is positive in aggregate but crosses zero at one density level, and the qualitative counterexamples in Section 3.3 include clear minimum-turn wins, so the evidence is conditional on this generator, configuration, comparator and density range rather than a general superiority claim. The representation audit is scientifically important for the same reason: it shows that scoring the rendered Stage-4 output instead of the Stage-3 centreline changes the result, so grouping, postprocessing and width representation must be kept separate.

Supervisor review note: Detailed discussion text intentionally omitted from this simplified review draft; the full discussion will be restored after supervisor feedback.

4.2 Development findings and failure modes

Development-only studies motivated three configuration choices without themselves being additional evidence: geometric and legacy curvature performed almost identically, so the fit-quality gate was left disabled; the one-pixel input mask outperformed wider variants; and a 2-by-2 grouping study found no hybrid design better than the current Hough-plus-FilaSeg combination. Residual errors have two directions, incorrect joins (over-merging) and refused continuations

(under-merging), and are hardest to resolve exactly at crossings, where tip geometry alone can be ambiguous.

Supervisor review note: Detailed discussion text intentionally omitted from this simplified review draft; the full discussion will be restored after supervisor feedback.

4.3 Scope and limitations

One real development crop. Every real-data result comes from one 512×512 crop from one parent image, and that crop informed configuration development. It is a case study, not a held-out estimate. A useful next study requires independently annotated crops from multiple parent images, instruments, materials and operating conditions, sealed until configuration lock.

Incomplete acquisition and annotation records. The repository does not establish the SEM instrument and acquisition settings or the full annotation protocol. These facts are marked as action required rather than inferred. They must be supplied before the manuscript can be submitted.

Synthetic transfer. The procedural generator exposes density, gaps, fading and crossing ambiguity under exact instance ground truth, but similarity of these distributions to real segmenter errors has not been quantified. The predefined confidence intervals characterize repeatability over generator seeds, not domain transfer to unseen SEM images. Prescribed density is thick-mask areal coverage, which jointly depends on centreline length and bundle width; the development factorial study therefore varies those factors independently and treats width as a Stage-3 negative control.

Input dependence. FilaSeg cannot recover a filament absent from the input centreline mask. Conversely, false axes may be retained or joined. The predefined comparison controls this dependency by supplying the identical mask to every method, but it does not remove it from practical use.

Representation and measurement. The common-fragment grouping metric is designed for instance identity, not boundary or width accuracy. The current real evidence does not support a strong per-instance width-accuracy claim, so such a claim is not made. Apparent width and other exported geometric quantities should be validated separately with matched per-instance measurements.

Comparator scope. The minimum-turn tracer is a transparent same-mask comparator, not a survey of the full tracing literature. Connected components are only a sanity check, and DeFiNe could not be run without introducing an unvalidated graph-construction adapter. The present data therefore support only a comparison with the fixed minimum-turn implementation used here.

Two-dimensional geometry. All inputs are two-dimensional grayscale SEM images. The output represents projected centrelines and overlapping instance layers; it does not recover three-dimensional over-and-under topology.

Reproducible release. The working repository records commands, environment versions, configuration hashes, seeds and machine-readable results. A citable archived release identifier and an independent clean-checkout reproduction remain required before submission.

5 Conclusions

FilaSeg reconstructs overlapping filament instances from degraded centreline masks through deterministic Hough vectorization, orientation-aware cleaning and geometric reconnection. On 125 predefined procedural scenes, its Stage-3 centreline output achieved common-fragment F_1 of 0.734 (density-stratified 95% CI [0.722, 0.745]). The paired difference from the fixed minimum-turn comparator was 0.071 (95% CI [0.059, 0.082]).

The evaluator audit changed the scientific interpretation without changing any frozen method output. Equivalent Stage-3 centrelines are the defensible primary grouping representation; Stage-4 centreline changes and width-rendered masks are secondary diagnostics. Automatically selected counterexamples show that the aggregate advantage is not uniform across scenes or densities. The result is therefore evidence for this fixed comparison under the tested generator, not a claim of superiority over filament tracing methods generally.

The single real crop remains a development illustration. Independent real annotations from multiple parent images, complete acquisition and annotation metadata, separate validation of apparent width and other exported measurements, and a citable clean-checkout release are required before broader SEM or quantitative microstructure claims are justified.

CRedit author contribution statement

TO BE COMPLETED BY THE AUTHORS BEFORE SUBMISSION. [ACTION REQUIRED: confirm each author’s CRediT roles against actual contributions; do not alter the preserved author names or order when completing this statement.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The FilaSeg source code, the procedural synthetic benchmark generator, the evaluation subsystem, and the manually annotated real test crop are contained in the accompanying repository. Every numerical result reported here is reproducible from that repository using Appendix C together with the full reproducibility record distributed with the manuscript. **[ACTION REQUIRED: a journal will require a citable archived release (for example a Zenodo DOI or a tagged GitHub release) rather than a bare repository URL; and Section 4.3 notes that the evaluation subsystem must be committed and a clean checkout confirmed to reproduce the reported numbers before submission]**

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Supervisor review note: This simplified review draft removes six appendices from the canonical manuscript (locked configuration and provenance, the connected-component sanity check, the curvature/fit-quality development study, near-tie qualitative examples, and the development-only hybrid grouping factorial), keeping only the independent density-width factorial, the dataset-identity record and the reproduction commands below. All removed material is present in full in the canonical paper repository.

A Independent centreline-density and bundle-width factorial

The original generator stops on thick-mask areal coverage, which couples total centreline length and physical width. A separate development generator instead stops on summed visible centreline arc length. It generated a 3-by-3 grid of target length densities and physical bundle widths with four disjoint seeds per cell. For each seed and length-density block, the degraded one-pixel input, clean centreline and overlap-aware centreline ground truth were byte-identical across all three physical-width cells. Width therefore acts as a structural negative control for Stage 3 and as an active factor only for SEM rendering and Stage 4.

Table 2. Achieved properties of the 9-cell development factorial, with four seeded replicates per cell (36 scenes total). Length density is in px per px²; thick-mask coverage is the union-area fraction. Each entry is the mean of four seeds. Crossing density, gap count and all centreline quantities are invariant across width within a geometry block.

Target length density	Achieved length density	Crossings per Mpx	Mean gap count	Thick-mask coverage by width		
				6 px	11 px	16 px
0.020	0.0208	118.3	13.0	0.121	0.212	0.295
0.040	0.0407	497.8	49.2	0.222	0.372	0.494
0.060	0.0607	1126.3	97.5	0.310	0.496	0.635

Increasing centreline density raised crossing density and gap count, while increasing physical width at fixed centreline geometry raised thick-mask coverage. The report records the full achieved width distribution, one-pixel coverage, degraded-mask coverage, filament count, pairwise and unique crossing events, and gap-length summaries. Empirical Stage-3 execution was not repeated across the redundant width cells; the negative-control result here is exact artifact identity, not an estimated performance effect.

B Dataset identity

The development set contains 20 procedural geometries and the single real crop. The disjoint predefined set was generated with base seed 20260801 and 25 scenes at each prescribed coverage. For density $d \in \{20, 30, 40, 50, 60\}$ and zero-based index i , the seed was $20260801 + 100d + i$. Thus the original predefined ranges were 20262801–20262825, 20263801–20263825, 20264801–20264825, 20265801–20265825 and 20266801–20266825. If the predefined precision rule required extension at a density, indices beginning at 25 were appended without replacing an existing scene. Exact scene paths, seeds, input and ground-truth hashes, method outputs and failure states are preserved in `evaluation_manifest.csv`.

C Reproduction

The predefined comparison was executed from the repository root as:

```
python eval/runners/experiment_runner.py \
-samples input/synthetic_locked_v1 \
-out output/locked_evaluation/results.json \
-mask-name mask_w1.png \
-methods baseline_cc,baseline_skeleton \
-filaseg run \
-filaseg-root output/locked_evaluation/filaseg \
-reconnect-config 3.reconnect/reconnect_config.yaml \
-cc-min-area 1 -max-gap-px 28 -max-turn-deg 20 \
-configuration-locked
```

The manuscript numerical source and figures were generated with:

```
python paper/scripts/results/make_result_macros.py \
-locked-json output/locked_evaluation/results.json \
-representation-json \
  output/evaluation_corrections/\
  locked_v1_representation_audit_corrected.json \
-curvature-json output/development_curvature_study/curvature_study.json \
-real-ablation-json output/final_real_ablation/real_ablation_results.json \
-out-dir paper/results

python paper/scripts/figures/fig_locked_results.py \
-corrected-representation-json \
  output/evaluation_corrections/\
  locked_v1_representation_audit_corrected.json \
-curvature-json output/development_curvature_study/curvature_study.json

python paper/scripts/figures/fig_paired_locked_examples.py \
-audit-json \
  output/evaluation_corrections/\
  locked_v1_representation_audit_corrected.json \
-output-dir paper/figures
```

The exact environment, validation commands, hashes and expected artifact paths are listed in docs/REPRODUCIBILITY.md. A citable archived-release identifier must replace the working-repository reference at submission.

[ACTION REQUIRED: move this appendix to the target journal's supplementary format and add the final archive DOI or release identifier.]